

Nonlinear Phase Portraits and Uniqueness Barriers

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Definition: Planar autonomous system

Let $U \subseteq \mathbb{R}^2$ be open, and let $f, g : U \rightarrow \mathbb{R}$. A **planar autonomous system** is

$$\dot{x} = f(x, y), \quad \dot{y} = g(x, y),$$

where $(x, y) \in U$.

Notation: Vector form of a planar system

Write

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \quad \mathbf{x} = (x, y), \quad \mathbf{F} : U \rightarrow \mathbb{R}^2, \quad \mathbf{F}(x, y) = (f(x, y), g(x, y)).$$

Recall: Equilibrium of a planar system

A point $(x^*, y^*) \in U$ is an **equilibrium** if

$$f(x^*, y^*) = g(x^*, y^*) = 0.$$

Definition: Nullclines

The **x -nullcline** is

$$\{(x, y) \in U : f(x, y) = 0\}.$$

The **y -nullcline** is

$$\{(x, y) \in U : g(x, y) = 0\}.$$

Observation: Directions along nullclines

Away from an equilibrium, the vector field is vertical on the x -nullcline and horizontal on the y -nullcline.

Observation: A nullcline need not be invariant

The condition $f(x, y) = 0$ makes the vector field vertical on the x -nullcline, but it does not guarantee that the vector field is tangent to that nullcline. Thus an x -nullcline need not be invariant. The analogous statement holds for a y -nullcline.

Recall: Invariant set

Let Φ be a possibly local flow. A set M is **invariant** if every solution starting in M remains in M throughout its maximal existence interval.

Definition: Positively invariant set

A set M is **positively invariant** if every solution starting at $\mathbf{x}_0 \in M$ satisfies

$$\Phi(t, \mathbf{x}_0) \in M$$

for every $t \geq 0$ in its maximal existence interval.

Question: What can nullclines determine before integration?

Analyze

$$\dot{x} = x + e^{-y}, \quad \dot{y} = -y.$$

Solution. The nullclines are

$$x = -e^{-y}, \quad y = 0.$$

Their unique intersection is the equilibrium $(-1, 0)$. The component signs are

region	sign $\dot{x}$	sign $\dot{y}$
$x > -e^{-y}, y > 0$	+	−
$x < -e^{-y}, y > 0$	−	−
$x > -e^{-y}, y < 0$	+	+
$x < -e^{-y}, y < 0$	−	+

The line $y = 0$ is invariant because $\dot{y} = 0$ there. On that line,

$$\dot{x} = x + 1,$$

so solutions move away from $(-1, 0)$. Also,

$$y(t) = y_0 e^{-t} \rightarrow 0 \quad (t \rightarrow \infty).$$

Thus $(-1, 0)$ repels trajectories along the invariant line $y = 0$. On the other hand, $y(t) \rightarrow 0$ as $t \rightarrow \infty$ for every initial value, so trajectories are attracted toward the line $y = 0$ in the y -coordinate.

Visualization: Nullcline sign map for the worked nonlinear system

	$x < -e^{-y}$	$x > -e^{-y}$
$y > 0$		
$y < 0$		

On $x = -e^{-y}$,

$$y > 0 : \downarrow, \quad y < 0 : \uparrow.$$

On $y = 0$,

$$x < -1 : \leftarrow, \quad x > -1 : \rightarrow.$$

Away from $x = -e^{-y}$, trajectory slopes satisfy

$$\frac{dy}{dx} = \frac{-y}{x + e^{-y}}.$$

Definition: Direction field

A **direction field** records the direction of $\mathbf{F}(\mathbf{x})$ at each state where $\mathbf{F}(\mathbf{x}) \neq \mathbf{0}$, but suppresses its magnitude.

Recall: Classical fourth-order Runge–Kutta method

Let $\mathbf{x}_n$ approximate a solution of $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ at time t_n , and let $h > 0$. The **classical fourth-order Runge–Kutta method** defines

$$\mathbf{k}_1 = \mathbf{F}(\mathbf{x}_n),$$

$$\mathbf{k}_2 = \mathbf{F}(\mathbf{x}_n + \frac{h}{2}\mathbf{k}_1),$$

$$\mathbf{k}_3 = \mathbf{F}(\mathbf{x}_n + \frac{h}{2}\mathbf{k}_2),$$

$$\mathbf{k}_4 = \mathbf{F}(\mathbf{x}_n + h\mathbf{k}_3),$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4).$$

Theorem: Local existence and uniqueness

Let $D \subseteq \mathbb{R}^n$ be open and let $\mathbf{F} : D \rightarrow \mathbb{R}^n$ be continuously differentiable. For every $\mathbf{x}_0 \in D$, the initial-value problem

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \quad \mathbf{x}(0) = \mathbf{x}_0,$$

has a unique solution on some open interval containing 0.

Corollary: Distinct maximal trajectories do not cross

Under the hypotheses of the local existence-and-uniqueness theorem, let $\mathbf{x}_j : I_j \rightarrow D$, $j = 1, 2$, be maximal solutions, meaning that neither solution extends to a larger time interval. Their trajectory images are either disjoint or identical.

Proof. Suppose

$$\mathbf{x}_1(t_1) = \mathbf{x}_2(t_2) = \mathbf{p}.$$

Define

$$\mathbf{y}_1(s) = \mathbf{x}_1(t_1 + s), \quad \mathbf{y}_2(s) = \mathbf{x}_2(t_2 + s).$$

Autonomy makes both solutions pass through $\mathbf{p}$ at time 0. Uniqueness makes them agree wherever both are defined, so they join to a solution on the union of their shifted time intervals. Maximality forces those intervals to be equal. Thus the original solutions trace the same trajectory, with different time origins.

Recall: Gradient

For a continuously differentiable function $h(x, y)$, its **gradient** is

$$\nabla h = (h_x, h_y).$$

Recall: Derivative along a solution

Let h be continuously differentiable on an open set containing a solution of $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$. The chain rule gives

$$\frac{d}{dt}h(x(t), y(t)) = h_x f + h_y g = \nabla h \cdot (f, g).$$

Proposition: Regular invariant-curve test

Let $D \subseteq \mathbb{R}^2$ be open, let $\mathbf{F} : D \rightarrow \mathbb{R}^2$ be locally Lipschitz, and let

$$C = \{(x, y) \in D : h(x, y) = 0\},$$

where $h \in C^1(D)$ and $\nabla h \neq \mathbf{0}$ on C . Suppose

$$\nabla h(x, y) \cdot \mathbf{F}(x, y) = 0 \quad \text{for every } (x, y) \in C.$$

Then C is invariant under $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$.

Proof. Near any point of C , the implicit-function theorem writes the curve as $y = k(x)$, after interchanging coordinates if needed. Write $\mathbf{F} = (f, g)$. Tangency gives

$$k'(x)f(x, k(x)) = g(x, k(x)).$$

Since k is C^1 , the function $x \mapsto f(x, k(x))$ is locally Lipschitz. Solve the scalar equation $\dot{x} = f(x, k(x))$ and set $y = k(x)$. The displayed identity gives $\dot{y} = g(x, y)$, so this is a solution of the full system lying on C . Uniqueness makes it the solution through the given point. This argument applies at every point of C , so a solution cannot leave C while it remains in D .

Question: Can a trajectory cross an invariant circle?

Consider

$$\dot{x} = -y + x(1 - x^2 - y^2), \quad \dot{y} = x + y(1 - x^2 - y^2).$$

Find the polar equations and classify the unit circle.

Solution. Let $r^2 = x^2 + y^2$. Then

$$\begin{aligned} \frac{d}{dt}r^2 &= 2x\dot{x} + 2y\dot{y} \\ &= 2(x^2 + y^2)(1 - x^2 - y^2) \\ &= 2r^2(1 - r^2). \end{aligned}$$

For $r > 0$,

$$\dot{r} = r(1 - r^2).$$

Also,

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = 1.$$

Hence

$$0 < r < 1 \Rightarrow \dot{r} > 0, \quad r = 1 \Rightarrow \dot{r} = 0, \quad r > 1 \Rightarrow \dot{r} < 0.$$

The unit circle is an invariant periodic orbit of period 2π . For $r_0 > 0$, uniqueness keeps $r(t)$ between r_0 and 1; compact containment gives existence for all $t \geq 0$. The radius is monotone, and its limit must be the only positive radial equilibrium, $r = 1$. Thus trajectories spiral outward toward the circle from inside and inward toward it from outside, without crossing it.

Proposition: A closed orbit is a barrier

Let C be a simple closed trajectory of a planar system with unique solutions. A trajectory beginning in the bounded component of $\mathbb{R}^2 \setminus C$ cannot enter the unbounded component while it exists.

Proof. By the Jordan curve theorem, every continuous path from the bounded component to the unbounded component meets C . Such an intersection would make the new trajectory coincide with C by uniqueness, which is impossible for a trajectory that began off C .

Question: Does local existence imply existence for all future time?

Solve

$$\dot{x} = x^2, \quad x(0) = x_0 > 0.$$

Solution. Separation gives

$$-\frac{1}{x(t)} = t + C.$$

The initial condition yields $C = -1/x_0$, so

$$x(t) = \frac{x_0}{1 - x_0 t}.$$

The right-hand side of the ODE is smooth on $\mathbb{R}$, but the solution becomes unbounded as $t \uparrow 1/x_0$. Thus local uniqueness does not imply global existence.

Theorem: Continuation from compact containment

Let $\mathbf{F} : D \rightarrow \mathbb{R}^n$ be locally Lipschitz on an open set D , and let $\mathbf{x} : [0, T_{\max}) \rightarrow D$ be a maximal solution. If $\mathbf{x}([0, T_{\max}))$ lies in a compact subset of D , then $T_{\max} = \infty$.

Equivalently, if $T_{\max} < \infty$, then for every compact set $K \subseteq D$ there exists $t_K < T_{\max}$ such that

$$\mathbf{x}(t) \notin K \quad \text{whenever } t_K < t < T_{\max}.$$

Idea: Analytical and numerical phase portraits

Analytically justified phase-portrait information often comes from

$$\boxed{\text{equilibria}} + \boxed{\text{nullclines and signs}} + \boxed{\text{invariant sets}} + \boxed{\text{uniqueness barriers}}.$$

Observation: A numerical trajectory is not a proof

A numerical trajectory is an approximation depending on its initial state, time interval, method, and step size. It can reveal a candidate invariant boundary or closed orbit, but it does not prove invariance, nonintersection, periodicity, or an asymptotic limit.